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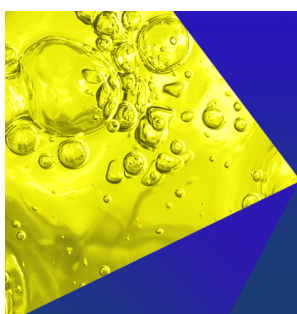
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ABSTRACT

For direct and large-eddy simulations of turbulent boundary layers (TBLs), the inflow turbulence conditions decisively influence the downstream flows. Recently, Wang *et al.* ["Inflow turbulence generation using an equivalent boundary layer model," Phys. Fluids **35**, 075110 (2023)] proposed an equivalent boundary layer (EBL) model to generate inflow turbulence fluctuations for zero-pressure gradient (ZPG) TBLs. In the present study, this model has been extended to TBLs with adverse pressure gradient (APG). Compared to the former model, the present model incorporates the mean wall-normal velocity, which is more prominent in APG TBLs than in ZPG TBLs. In addition, a filter is imposed on the streamwise velocity to suppress the amplified streamwise large-scale structures. The driving force was obtained from the databases and the prediction models of the APG TBLs. The present EBL model was applied to the generation of the inflow turbulence for two APG TBLs with $(Re_\theta, \beta) = (2180, 2.2)$ and $(4100, 1.4)$ at the inlet, where Re_θ is the Reynolds number based on the momentum thickness θ and β is the Rotta–Clauser pressure-gradient parameter. The results show that the EBL model well reconstructs the turbulent fluctuations of the TBL at the corresponding Re_θ and β and can ensure the natural downstream development of the flow when used as the inflow for streamwise developing TBL simulations.

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I. INTRODUCTION

Turbulent boundary layers with zero pressure gradient (ZPG) are a kind of well-studied canonical wall-bounded turbulent flows. However, TBLs with non-zero pressure gradients are more common and significant in engineering applications, such as flows over curved surfaces and with attack angles. Clauser *et al.*¹ proposed the pressure gradient parameter, $\beta = (\delta^*/\tau_w)(dP/dx)$, where δ^* is the displacement thickness, τ_w is the wall shear stress and dP/dx is the far-field streamwise pressure gradient. If $\beta > 0$, the TBL subjects to APG effects, whereas, if $\beta < 0$, it is exposed to the influence of favorable pressure gradient (FPG). The TBL is called in equilibrium state if β is a constant independent of streamwise position.

The statistical and structural characteristics of the equilibrium APG TBLs have been studied by experiments^{1–5} and numerical simulations,^{6–10} just to name a few. The APG enhances the wake region and results in a secondary peak of streamwise Reynolds stress $\langle u'u' \rangle$ in the outer layer.⁶ It intensifies vortex motion in the outer region and sweep events in the near-wall region.⁶ Bobke *et al.*⁸ showed that the

turbulent statistics of APG TBLs are determined by the friction Reynolds number Re_τ , the pressure-gradient parameter β , and its upstream history. The history effect of β makes the inlet boundary conditions impose significant influence on the downstream development of the TBL. An efficient method to give the inlet turbulence for APG TBLs is still open for investigation.

The existing methods to generate the inlet BCs for APG TBL simulations can be categorized into three groups. In the first group, the inlet boundary is set in the laminar region and turbulent flow is obtained via transition triggered by adding disturbances. The method with trip at a short distance downstream of the inlet was used for the simulation of APG TBLs using spectral methods.^{7,8} To use periodic boundary conditions in the streamwise direction, a fringe region was set upstream of the outlet, which forced the developed turbulent flow back to the laminar inflow. In the study of Skote and Henningson,⁷ the transition happened at approximately $x/\delta_0^* = 150$, where δ_0^* is the displacement thickness of the laminar inflow. The flow consumed 3/8 of the entire computational domain to be self-similar in the inner scaling.

Jacobs and Durbin¹¹ proposed a different approach by imposing a superposition of the Blasius velocity profile and isotropic free-stream turbulence as the inlet BCs. This method was employed by Yoon *et al.*,¹² and the distribution of the skin friction coefficient (C_f) indicated that nearly half of the computational domain was necessary for the flow to reach equilibrium.

In the second group, the inlet boundary is set in the ZPG TBL region. The recycling-rescaling method proposed by Lund *et al.*¹³ is a commonly adopted method for the auxiliary simulation of ZPG TBLs. This method has been used by Lee and Sung¹⁴ and Lee,⁶ whose auxiliary simulations were both ZPG TBL flows at $Re_\theta = 300$. In Lee's study,⁶ approximately one-third of the computational domain was required to achieve a constant β . A previous DNS of ZPG TBL¹⁵ was used by Rkein and Laval¹⁶ to be the inlet BCs of APG TBL. The precursor DNS had smaller computational domain so the inlet velocity fields were extrapolated in the wall-normal direction and periodized in the spanwise direction. To avoid the exact repetition in the results because of the duplication of inlet planes, a spatial-temporal noise of small amplitude was added to the inlet velocity components. Despite these advancements, the computational cost associated with these methods remains significant. Sillero *et al.*¹⁷ proposed a modified recycling method of Lund *et al.*,¹³ in which a mapping depending on wall-normal position was used to rescale the velocities from the reference plane to the inflow. Kitsios *et al.*¹⁸ adopted this approach and set the reference plane in the ZPG region. However, their TBLs only reached self-similarity approximately $2000\delta_{in}^*$ (δ_{in}^* is the displacement thickness at the inlet) downstream of the inlet, consuming nearly half of the computational domain.

In the third group, APG TBL BCs were prescribed directly at the inlet. Using the modified recycling method¹⁷ and setting the reference plane in the APG region, Kitsios *et al.*¹⁹ performed DNS to a self-similar APG TBL at the verge of separation. The wall-normal velocity was prescribed from the inlet to impose the APG, and the initial inlet profiles had the spanwise homogeneous Fourier mode rescaled and interpolated from the time-averaged profiles of a previously simulated APG TBL. However, their domains of interest, which were in a self-similar state, were around 1/10 of the whole computational domain. There still lacks of a more efficient technique to directly generate inlet turbulence for APG TBLs.

Recently, Wang *et al.*²⁰ proposed the equivalent boundary layer (EBL) model and applied it to the generation of inflow turbulence for ZPG TBLs. In the EBL model, the slowly streamwise developing TBL at a given Reynolds number is approximated by a streamwise homogeneous wall-bounded turbulent flow by adding an equivalent body force to the streamwise momentum equation. Therefore, the flow in EBL obeys the same mean momentum balance as in TBL. The EBL model has several advantages in generating inflow turbulence for TBL. It requires no self-similar laws of TBL and is easily extended to TBLs subjected to some complex effects, such as compressibility,²¹ wall roughness, and pressure gradient. It is statistically homogeneous and allows for the use of periodic conditions in the streamwise direction. The inflow turbulence generated by EBL is more accurate and shortens the adjustment length after the inlet.

The EBL model proposed by Wang *et al.*²⁰ is only used in ZPG TBLs. In the present study, the model is improved to generate inflow turbulence for APG TBLs. Section II introduces the mathematical formulation and the numerical method for the direct numerical

simulation of EBL and TBL. Results of EBL and TBL are presented in Sec. III. Section IV gives the conclusion.

II. MATHEMATICAL FORMULATION

A. EBL model

The governing equations of the present EBL model are as follows:

$$\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial p}{\partial x_i} + \nu \frac{\partial^2 u_i}{\partial x_j \partial x_j} + F \delta_{i1}, \quad (1a)$$

$$\frac{\partial u_i}{\partial x_i} = M, \quad (1b)$$

in which x_1, x_2, x_3 or x, y, z represent the streamwise, wall-normal, and spanwise directions, and u_1, u_2, u_3 or u, v, w are the corresponding velocity components. p, ρ and ν are the pressure, density, and kinematic viscosity of the fluid. F and M are source terms to ensure the EBL follows the same streamwise mean momentum equation and continuity equation as the TBL at the corresponding Reynolds number.²⁰ They satisfy

$$F = -\frac{\partial \langle u'v' \rangle_{tbl}}{\partial y} + \nu \frac{\partial^2 U_{tbl}}{\partial y^2} - V \frac{\partial U}{\partial y} \quad (2)$$

and

$$M = \frac{\partial V_{tbl}}{\partial y}, \quad (3)$$

in which U and V are the mean streamwise and wall-normal velocities, and $\langle u'v' \rangle$ is the Reynolds shear stress. $\langle \cdot \rangle$ denotes the average with the spanwise direction and time. The terms with subscript “tbl” represent the quantities of the equilibrium APG TBL at the same Re_θ and β as the EBL. V_{tbl} can be obtained from the priori databases of APG TBLs or the prediction model proposed by Wei *et al.*²²

The EBL flow is statistically homogeneous in the streamwise and spanwise directions, therefore the periodic boundary conditions are used in the x - and z -directions. No-slip conditions are applied at the wall, and Neumann conditions are imposed at the top boundary as

$$y = h : \frac{\partial u}{\partial y} = 0, \quad \frac{\partial v}{\partial y} = \frac{\partial V_{tbl}}{\partial y}, \quad \frac{\partial w}{\partial y} = 0, \quad (4)$$

where h is the height of the EBL model.

Based on the original model proposed by Wang *et al.*,²⁰ the present EBL model further incorporated the influence of wall-normal velocity into the formulation. Wei *et al.*²² showed that the mean wall-normal velocity under APG is significantly larger than that at or near ZPG. According to the transport of the turbulent kinetic energy, the convection term is $U \partial k / \partial x + V \partial k / \partial y$, where k is the turbulent kinetic energy. The $V \partial k / \partial y$ promotes the near-wall turbulent kinetic transfer to the upper layer, helping reconstruct the large-scale structure in the outer layer. To embody the wall-normal convection features of TBL, we incorporated a mass source term M into the EBL model, as shown in Eq. (1b). To ensure momentum conservation, we introduced the term $-V \partial U / \partial y$ into the body force term as shown in Eq. (2). Consequently, the no-penetration condition at the top boundary is replaced by the Neumann condition as shown in Eq. (4).

Another feature of APG TBL, in comparison with ZPG TBL, is the prominent large-scale motions present in the outer region, resulting in the second peak in $\langle u'u' \rangle$ and the spanwise premultiplied energy spectrum. In EBL, due to the periodic conditions in the streamwise direction, these large-scale structures would be reinforced, leading to the nonphysical large-scale motions in the streamwise direction. To alleviate the influence of these nonphysical structures which are difficult to dissipate downstream, we removed the streamwise uniform fluctuations from the streamwise velocity component when the EBL turbulence was used as the inlet conditions for the simulation of APG TBL. The following statistics show that the physical streamwise large-scale structures emerge after the inlet of TBLs.

To validate the performance of the EBL model in the absence of errors introduced by the source terms, the DNS data of previous studies were first used to calculate F and M , as shown in Fig. 1. The wall-normal distributions of F at $(Re_\theta, \beta) = (2180, 2.2)$ from Lee⁵ and at $(Re_\theta, \beta) = (4100, 1.4)$ from Yoon *et al.*¹² are shown in Fig. 1 by black solid and dashed lines, respectively. Very recently, Shu and Xu²³ proposed a method to predict the profiles of mean velocity and total shear stress of APG TBL. The distributions of F given by the predictive

model are also shown in Fig. 1 by the red lines. They agree quite well with the DNS results.

B. APG TBL

The governing equations for the streamwise developing APG TBLs can also be written as Eq. (1) by setting the source terms $F = 0$ and $M = 0$. For the boundary conditions, at inlet, the mean velocity obtained from the DNS database and turbulence fluctuations generated by the corresponding EBL simulation were prescribed. At the outlet, the modified convective boundary condition $\partial \mathbf{u} / \partial t + C \partial \mathbf{u} / \partial x = C \partial \mathbf{U} / \partial x$ proposed by Wang *et al.*²⁰ was adopted, where C is the convection speed and $\mathbf{U}$ is the mean velocity at the outlet. This improved convective condition can ensure the natural development of the flow near the outlet. Periodic conditions were applied in the spanwise direction, and no-slip conditions at the wall. At the top boundary, a power-law distribution of the free-stream velocity was assigned to the streamwise velocity $u = U_\infty(x) \sim x^m$ to impose the pressure gradient, and the Neumann boundary conditions were adopted for wall-normal and spanwise velocity components as $\partial v / \partial y = -\partial U_\infty / \partial x$, $\partial w / \partial y = 0$.

C. Numerical methods

In this study, the governing equations (1) are solved by the second-order central difference scheme and the Crank–Nicolson scheme for spatial and temporal discretization, respectively. The continuity and momentum equations are decoupled by the projection method proposed by Kim *et al.*,²⁴ which can ensure the second-order precision in both space and time. The numerical method was first proposed by Perot²⁵ and Kim *et al.*²⁴ and has been validated in our previous research Huang *et al.*²⁶ and Wang *et al.*²⁰

III. RESULTS

A. EBL model

DNSs of the EBL flows were carried out under two different settings $(Re_\theta, \beta) = (2180, 2.2)$ and $(4100, 1.4)$, which were used for generating inflow fluctuations of two corresponding equilibrium APG TBLs. The computational parameters are shown in Table I, in which $Re = U_\infty \delta / \nu$ and $Re_\tau = u_\tau \delta / \nu$. δ is the boundary layer thickness and $u_\tau = (\tau_w / \rho)^{0.5}$. The EBL simulations are initiated from the fully developed turbulent open channel flows at similar Re_τ . Cases EBL-2180 and EBL-4100 use source terms calculated from DNS data, and the cases using the predictive models²³ are named by EBL-2180-M and EBL-4100-M, respectively.

The turbulent statistics of EBLs are compared with the DNS data of the streamwise developing APG TBLs at the corresponding

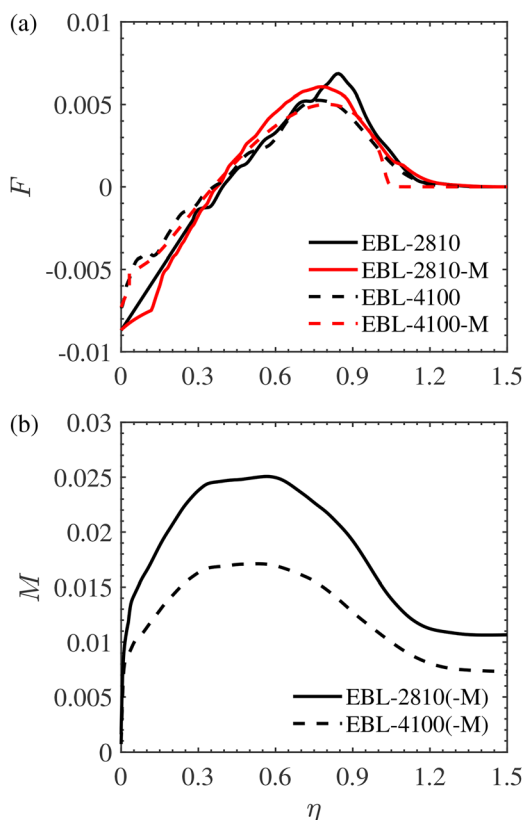


FIG. 1. Wall-normal distributions of (a) the body force F and (b) the mass source term M . In EBL-2180 and EBL-4100, F is calculated from the DNS data of Lee⁵ and Yoon *et al.*,¹² respectively; in EBL-2180-M and EBL-4100-M, F is given by the predictive model.²³ M is calculated using the mean wall-normal velocity from the predictive model of Wei *et al.*²² F and M is normalized by U_∞^2 / δ and U_∞ / δ , respectively, and $\eta = y / \delta$.

TABLE I. Parameters of the EBL cases. The case names are numbered by Re_θ of their corresponding TBLs' inlet.

Case name	EBL-2180	EBL-4100
Re	8927	19602
Re_τ	346	690
$(L_x, L_y, L_z) / \delta$	12.56, 6, 6.28	8, 4, 4
N_x, N_y, N_z	512, 512, 512	512, 450, 512
$\Delta x^+, \Delta y_{min}^+, \Delta y_e^+, \Delta z^+$	8.5, 0.10, 4.8, 4.3	10.7, 0.07, 9.7, 5.4

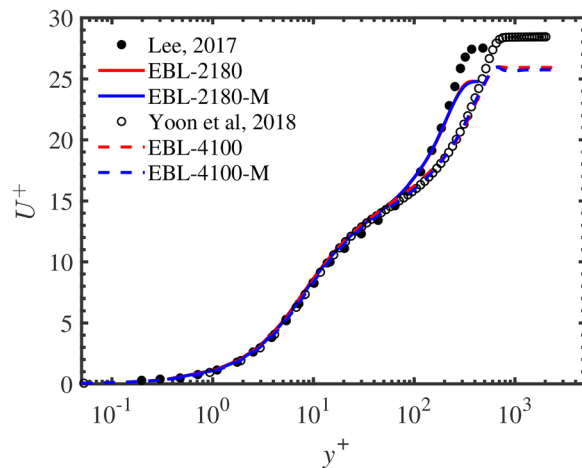


FIG. 2. Mean velocity profiles.

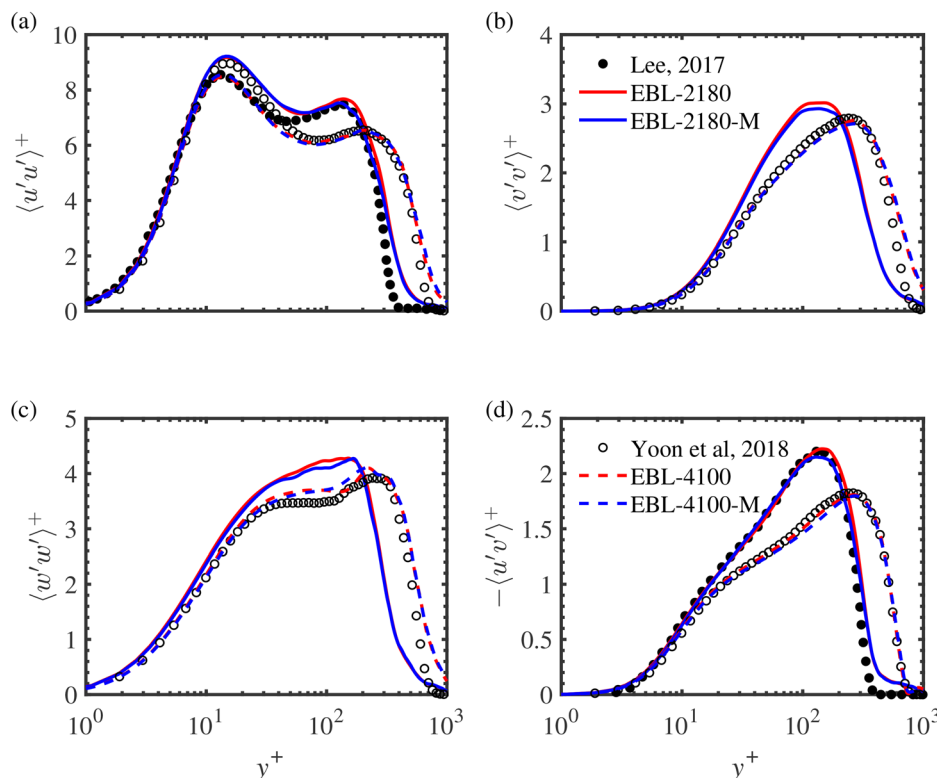
Reynolds numbers and pressure gradient parameters.^{6,14} Figure 2 shows the mean velocity profiles for cases EBL-2180 and EBL-4100. The superscript + denote the normalization by the inner scale. The mean velocity profile collapsed well in $y^+ < 200$ for case EBL-2180 and in $y^+ < 500$ for case EBL-4100, implying that the EBL model can reconstruct the near-wall motions. In the wake region, the mean velocity of TBL is higher than the EBL model for both cases. As demonstrated by Lee,⁶ Harun *et al.*³ and Lee and Sung,¹⁴ the wake region of

the APG TBLs are enhanced compared to the ZPG TBLs at similar Re_τ primarily due to the increase in the turbulence intensity in the outer region. The results using the source terms given by the predictive models show no difference with those using DNS data.

Figure 3 presents the Reynolds stresses profiles. The EBL results collapse well with the TBL results. For $\langle u'u' \rangle^+$, the EBL model can accurately predict the second peak in the outer region, indicating that EBL is able to reproduce the large-scale structures present in the outer region of the APG TBL. For the $\langle w'w' \rangle$ in the case EBL-4100, the EBL model correctly captures both the plateau region and the position of the peak, demonstrating the model's ability to accurately characterize spanwise motions. The EBL model also successfully captures the wall-normal profile of $\langle u'v' \rangle$, which is closely related to the ejection and sweep events.

Figure 4 shows the root-mean square of the streamwise, wall-normal and spanwise vorticity fluctuations predicted by the EBL model. According to Yoon *et al.*,¹² compared to ZPG TBLs, the vorticity fluctuations in APG TBL are more active in the near-wall region ($y^+ < 15$) and have a shoulder in the outer region (around $y^+ = 400$). For both case EBL-2180 and EBL-4100, an increase in vorticity is observed in the outer region ($y^+ > 100$). In the case EBL-2180, the shoulders of $\omega_{x,rms}^+$ and $\omega_{y,rms}^+$ appear near $y^+ = 200$. For case EBL-4100, the shoulders of $\omega_{x,rms}^+$ and $\omega_{y,rms}^+$ appear near $y^+ = 300$. These are signatures of the large-scale motions that exist in the APG TBLs.

In aforementioned first- and second-order statistics, the EBL models using the source terms obtained from the DNS data or the predictive models give very close results and are both consistent with the statistics of authentic TBL. This extends the usability of the EBL model.

FIG. 3. Profiles of (a) $\langle u'u' \rangle^+$, (b) $\langle v'v' \rangle^+$, (c) $\langle w'w' \rangle^+$, and (d) $-\langle u'v' \rangle^+$.

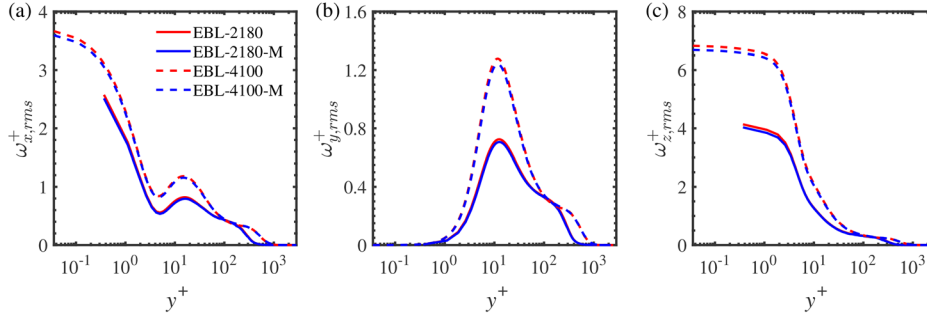


FIG. 4. Profiles of (a) $\omega_{x,rms}^{I+}$, (b) $\omega_{y,rms}^{I+}$, and (c) $\omega_{z,rms}^{I+}$.

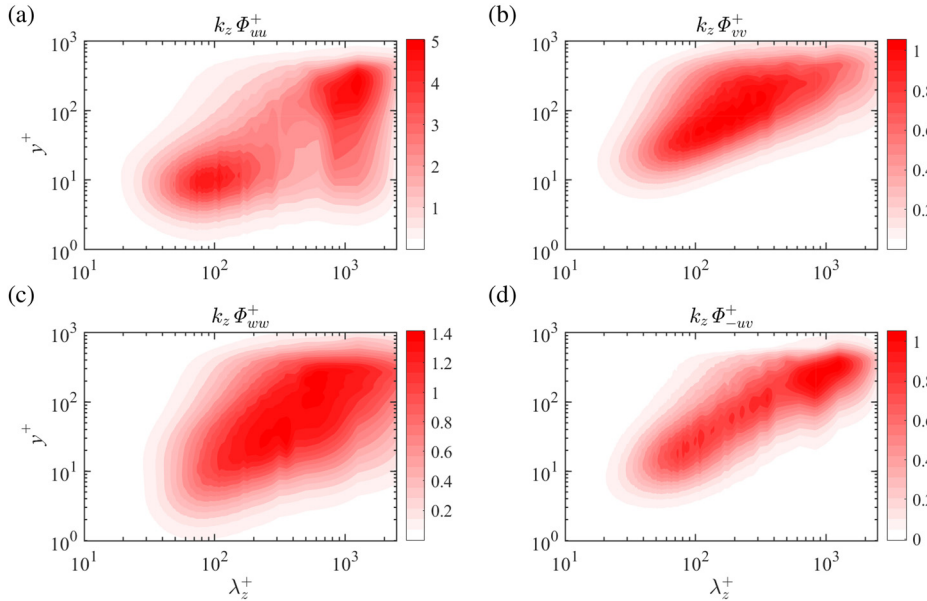


FIG. 5. Premultiplied spanwise energy spectra of (a) the streamwise, (b) the wall-normal, (c) the spanwise velocity fluctuations, and (d) the Reynolds shear stress.

When it's necessary to provide an inlet BC for the equilibrium APG TBLs with specific Re_τ and β , the predictive model can be used to generate the source terms. Then the EBL model can generate the inlet turbulence, which is combined with the mean streamwise velocity to form the inlet velocity BC.

Figure 5 displays the premultiplied spanwise energy spectra of the Reynolds stresses of EBL-4100. They are defined as follows:

$$k_z \Phi_{u_i u_i}^+(y^+, \lambda_z^+) = k_z \frac{\sum_{\lambda_x^+} E_{u_i u_i}(y^+, \lambda_x^+, \lambda_z^+)}{\sum_{y^+} \sum_{\lambda_x^+} E_{u_i u_i}(y^+, \lambda_x^+, \lambda_z^+)}, \quad (5a)$$

$$E_{u_i u_i}(y^+, \lambda_x^+, \lambda_z^+) = \hat{u}_i(y^+, \lambda_x^+, \lambda_z^+) \hat{u}_i^*(y^+, \lambda_x^+, \lambda_z^+), \quad (5b)$$

where $\hat{u}_i$ is the Fourier transform of velocity u_i with respect to x and z , and $\hat{u}_i^*$ is the complex conjugate of $\hat{u}_i$. λ_x^+, λ_z^+ are the wavelength in the streamwise and spanwise direction. The premultiplied spanwise energy spectra in APG TBLs¹² indicated that $k_z \Phi_{uu}^+$ and $k_z \Phi_{-uv}^+$ has two peaks located in the inner and outer regions, respectively. The outer peak is caused by the large-scale structures in the outer region. This phenomenon also occurs in EBL-4100 with peaks located at $y^+ = 10$ and $y^+ = 300$. Furthermore, in the EBL model, the outer peak of

$k_z \Phi_{-uv}^+$ originates from the near-wall region and extends to the outer layer, consistent with the results obtained by Lee⁶ and Yoon *et al.*¹²

B. TBL

The EBL model was applied to generate the inlet fluctuations for the DNS of streamwise developing APG TBL. The computational parameters are shown in Table II. Both cases are applied the top BCs

TABLE II. Parameters of the TBL cases. The case names are numbered by Re_θ at the inlet.

Case name	TBL-2180	TBL-4100
Auxiliary simulation	EBL-2180	EBL-4100
Re_{in}	8927	19602
Re_θ	2180–4000	4100–6000
Re_τ	288–346	690–900
$(L_x, L_y, L_z)/\delta_{in}$	25.12, 6, 6.28	24.0, 4.0, 4.0
N_x, N_y, N_z	1024, 512, 512	1536, 450, 512
$\Delta x^+, \Delta y_{min}^+, \Delta y_e^+, \Delta z^+$	8.5, 0.10, 4.8, 4.3	10.7, 0.07, 3.2, 5.4

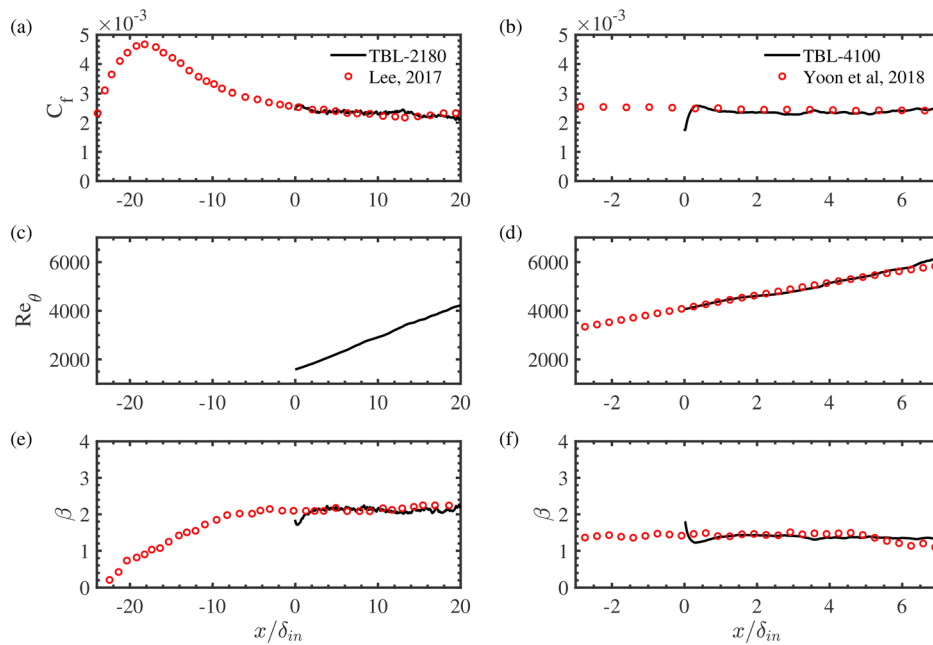


FIG. 6. Streamwise developing statistics of TBL-2180 and TBL-4100, varying with x/δ_{in} : (a,b) friction coefficient C_f , (c,d) Re_θ , and (e,f) Clauser pressure gradient parameter β .

$u = U_\infty(x) \sim x^m$. For TBL-2180 the constant $m = -0.25$ was used to impose the PG with $\beta = 2.2$ and for TBL-4100 $m = -0.2$ to impose the PG with $\beta = 1.4$.

First, the streamwise distributions of skin friction coefficient C_f , the Reynolds number based on momentum thickness Re_θ and pressure-gradient parameter β are compared with the DNS data of Lee⁶ and Yoon *et al.*,¹² as shown in Fig. 6. The $x = 0$ denotes the inlet location of the computational domains for case TBL-2180 and TBL-4100, and the location at which the mean velocity U and Reynolds shear stress $\langle u'v' \rangle$ are extracted from Lee,⁶ and Yoon *et al.*,¹² TBL to generate the EBL source terms. It is clear that the streamwise adjustment length is shorter than $2\delta_{in}$ for both TBL-2180 and TBL-4100 cases.

For the friction coefficient C_f shown in Figs. 6(a) and 6(b), both TBL-2180 and TBL-4100 exhibit no change in the equilibrium region, consistent with the results in the literature. The streamwise length to achieve equilibrium APG TBL of Kitsios,^{18,19} Lee,⁶ and Rkein *et al.*,¹⁶ Yoon *et al.*¹² is, respectively, $6/7$, $1/3$, and $1/2$ of the computational domain, nearly $10\text{--}20\delta_{in}$. However, the present C_f demonstrates that the EBL model significantly reduces the streamwise adjustment length to $1\text{--}2\delta_{in}$. In Fig. 6(a), TBL-2180 reaches equilibrium before $x = 10\delta_{in}$ and reduces the computational domain of $20\delta_{in}$ (compared to case of Lee⁶), reducing $2/3$ of CPU hours for computation. This suggests the efficiency of the present EBL-based method in inflow turbulence generation.

Figures 6(c) and 6(d) present the streamwise development of Re_θ of TBL-2180 and TBL-4100. The streamwise variation of Re_θ in TBL-4100 is consistent with the literature. This confirms that using the EBL method, the momentum loss from the inlet to the downstream region requires a very short adjustment region.

The streamwise variation of β also matches the literature data, remaining nearly constant as shown in Figs. 6(e) and 6(f). This indicates that the EBL model enables the boundary layer to rapidly reach equilibrium. In the DNS of Lee,⁶ with a ZPG inlet, β increases from 0

to the specified value, consuming nearly $1/3$ of the computational domain. Furthermore, in the initial region where β is constant, the turbulence statistics still vary along the streamwise direction due to the history effect. An effective equilibrium boundary layer is only achieved at $20\delta_{in}$ downstream of the inlet. For the present study, this adjustment region is reduced to one-fifth of the domain length.

The streamwise variation of the shape factor $H_{12} = \delta^*/\theta$ further verifies that the present TBL simulations rapidly achieve the equilibrium state as shown in Fig. 7(a). Figure 7(b) shows the streamwise evolution of the peak values of turbulence intensities, u'^+_{rms} , v'^+_{rms} , and w'^+_{rms} . The results indicate the flows naturally develop downstream from the

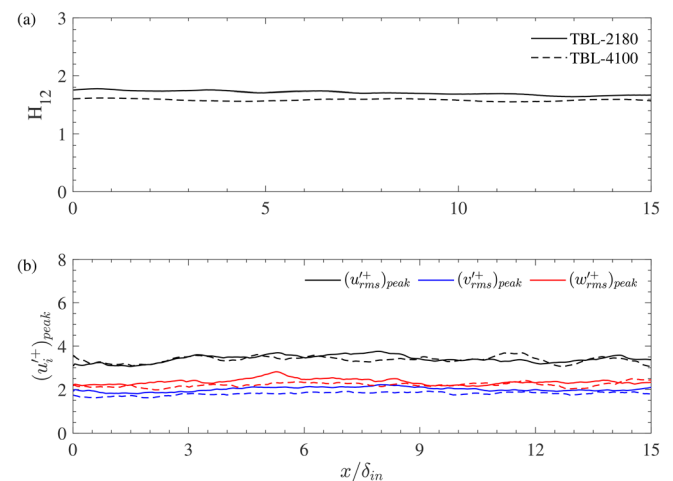


FIG. 7. Streamwise developing statistics of TBL-2180 (solid lines) and TBL-4100 (dashed lines). (a) The shape factor $H_{12} = \delta^*/\theta$. (b) Peak values of velocity fluctuations.

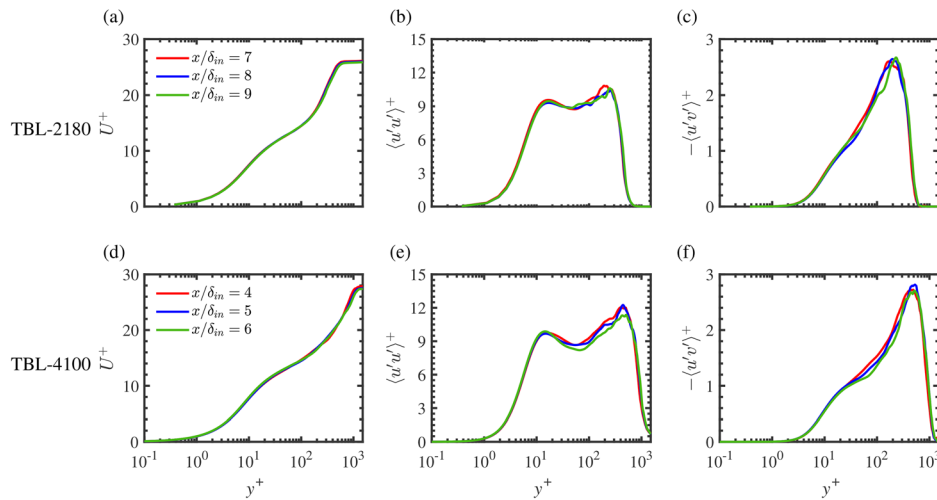


FIG. 8. Statistics in the equilibrium region of TBL-2180 and TBL-4100: (a, d) U^+ , (b, e) $\langle u'u' \rangle^+$, and (c, f) $-\langle u'v' \rangle^+$.

inlet with a very short adjustment region. Comparing the two cases, TBL-2180 and TBL-4100, we observed that although case TBL-2180 has a smaller Reynolds number, it has slightly larger turbulence intensities when normalized by friction velocity u_τ . This indicates that APG can enhance turbulence intensity, since the pressure gradient β is 2.2 for TBL-2180 and 1.4 for TBL-4100. The excitation effect of APG on turbulence is consistent with the former study. Hoyas *et al.*²⁷ and Bernardini *et al.*²⁸ demonstrated that boundary layers under APG exhibit flow characteristics similar to those at higher Reynolds numbers.

Figure 8 shows the profiles of U^+ , $\langle u'u' \rangle^+$, and $-\langle u'v' \rangle^+$ at three streamwise locations in the equilibrium region. Within the streamwise range of $3\delta_m$, both cases exhibit negligible variation when non-dimensionalised by inner scale. Notably, the second peak of $\langle u'u' \rangle^+$ is pronounced, demonstrating the strengthening effect of APG on outer-region turbulence fluctuations.

IV. CONCLUSION

In this paper, an efficient method is proposed to generate the inflow turbulence for the simulation of APG TBL. This method uses an equivalent boundary layer model, which approximate the streamwise slowly varying TBL with a streamwise homogeneous flow by adding source terms to the governing equations. The EBL model allows for the usage of periodic conditions in the streamwise direction. The source terms guarantee the EBL flow follows the mean momentum balance and mass conservation the same as TBL at the given streamwise location. The source terms can be calculated from the priori databases of APG TBL or by the prediction models. Compared with the original model proposed by Wang *et al.*,²⁰ the present EBL model considered the influence of the mean wall-normal velocity by adding the mass source and correcting the upper boundary condition. A filter is imposed to suppress the nonphysical structures.

By generating the inflow turbulence for the DNS of APG TBL with inlet $(Re_\theta, \beta) = (2180, 2.2)$ and $(4100, 1.4)$, the EBL model displayed the following three advantages. First, the EBL model well reconstructs the statistical characteristics of TBL at corresponding Re_θ and β . Second, the EBL model shortens the adjustment length of streamwise developing statistics. This method avoids the long adjustment

region from ZPG TBL to APG TBL needed by auxiliary ZPG simulation. Third, based on the total stress model of APG TBL, the EBL model can be used to generate inflow turbulence for equilibrium APG TBL at moderate Reynolds number and pressure gradient. For the non-equilibrium APG TBLs, the statistics change in the streamwise direction due to the history effect. The EBL model simulates a streamwise periodic flow to reconstruct turbulent boundary layers with slow streamwise development. The current EBL model is indeed incapable of generating inflow conditions for non-equilibrium flows. This is a critical challenge that needs to be addressed in future work.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Yi Li: Data curation (lead); Methodology (equal); Validation (lead); Writing – original draft (lead). **Zheng-Qin Shu:** Data curation (equal); Validation (supporting). **Wei-Xi Huang:** Supervision (equal); Writing – review & editing (supporting). **Chun-Xiao Xu:** Funding acquisition (lead); Methodology (equal); Supervision (lead); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

- ¹F. H. Clauser, "Turbulent boundary layers in adverse pressure gradients," *J. Aeronaut. Sci.* **21**, 91–108 (1954).

- ²J. Monty, Z. Harun, and I. Marusic, "A parametric study of adverse pressure gradient turbulent boundary layers," *Int. J. Heat Fluid Flow* **32**, 575–585 (2011).
- ³Z. Harun, J. P. Monty, R. Mathis, and I. Marusic, "Pressure gradient effects on the large-scale structure of turbulent boundary layers," *J. Fluid Mech.* **715**, 477–498 (2013).
- ⁴R. Deshpande, A. Van Den Bogaard, R. Vinuesa, L. Lindić, and I. Marusic, "Reynolds-number effects on the outer region of adverse-pressure-gradient turbulent boundary layers," *Phys. Rev. Fluids* **8**, 124604 (2023).
- ⁵P.-Å. Krogstad and P. E. Skåre, "Influence of a strong adverse pressure gradient on the turbulent structure in a boundary layer," *Phys. Fluids* **7**, 2014–2024 (1995).
- ⁶J. H. Lee, "Large-scale motions in turbulent boundary layers subjected to adverse pressure gradients," *J. Fluid Mech.* **810**, 323–361 (2017).
- ⁷M. Skote, D. S. Henningson, and R. A. Henkes, "Direct numerical simulation of self-similar turbulent boundary layers in adverse pressure gradients," *Flow, Turbul. Combust.* **60**, 47–85 (1998).
- ⁸A. Bobke, R. Vinuesa, R. Örlü, and P. Schlatter, "History effects and near equilibrium in adverse-pressure-gradient turbulent boundary layers," *J. Fluid Mech.* **820**, 667–692 (2017).
- ⁹M.-b. Sun, Z. Hu, and N. D. Sandham, "Recovery of a supersonic turbulent boundary layer after an expansion corner," *Phys. Fluids* **29**, 076103 (2017).
- ¹⁰M. Sun, N. D. Sandham, and Z. Hu, "Turbulence structures and statistics of a supersonic turbulent boundary layer subjected to concave surface curvature," *J. Fluid Mech.* **865**, 60–99 (2019).
- ¹¹R. G. JACOBS and P. A. DURBIN, "Simulations of bypass transition," *J. Fluid Mech.* **428**, 185–212 (2001).
- ¹²M. Yoon, J. Hwang, and H. J. Sung, "Contribution of large-scale motions to the skin friction in a moderate adverse pressure gradient turbulent boundary layer," *J. Fluid Mech.* **848**, 288–311 (2018).
- ¹³T. S. Lund, X. Wu, and K. D. Squires, "Generation of turbulent inflow data for spatially-developing boundary layer simulations," *J. Comput. Phys.* **140**, 233–258 (1998).
- ¹⁴J.-H. Lee and H. J. Sung, "Effects of an adverse pressure gradient on a turbulent boundary layer," *Int. J. Heat Fluid Flow* **29**, 568–578 (2008).
- ¹⁵I. Solak and J.-P. Laval, "Large-scale motions from a direct numerical simulation of a turbulent boundary layer," *Phys. Rev. E* **98**, 033101 (2018).
- ¹⁶H. Rkein and J.-P. Laval, "Direct numerical simulation of adverse pressure gradient turbulent boundary layer up to $Re_\theta \simeq 8000$," *Int. J. Heat Fluid Flow* **103**, 109195 (2023).
- ¹⁷J. A. Sillero, J. Jiménez, and R. D. Moser, "One-point statistics for turbulent wall-bounded flows at Reynolds numbers up to $\delta^+ \approx 2000$," *Phys. Fluids* **25**, 105102 (2013).
- ¹⁸V. Kitsios, C. Atkinson, J. Sillero, G. Borrell, A. Gungor, J. Jiménez, and J. Soria, "Direct numerical simulation of a self-similar adverse pressure gradient turbulent boundary layer," *Int. J. Heat Fluid Flow* **61**, 129–136 (2016).
- ¹⁹V. Kitsios, A. Sekimoto, C. Atkinson, J. A. Sillero, G. Borrell, A. G. Gungor, J. Jiménez, and J. Soria, "Direct numerical simulation of a self-similar adverse pressure gradient turbulent boundary layer at the verge of separation," *J. Fluid Mech.* **829**, 392–419 (2017).
- ²⁰H.-N. Wang, R.-X. Li, W.-X. Huang, and C.-X. Xu, "Inflow turbulence generation using an equivalent boundary layer model," *Phys. Fluids* **35**, 075110 (2023).
- ²¹R.-X. Li, W.-X. Huang, and C.-X. Xu, "Inflow turbulence generation for compressible turbulent boundary layers," *Phys. Fluids* **36**, 025132 (2024).
- ²²T. Wei, Z. Li, T. Knopp, and R. Vinuesa, "The mean wall-normal velocity in turbulent boundary layer flows under pressure gradient," *J. Fluid Mech.* **975**, A27 (2023).
- ²³Z. Shu and C. Xu, "Mean velocity profiles and total shear stress profiles in adverse-pressure-gradient turbulent boundary layers considering history effect," *arXiv:2508.07663* [physics.flu-dyn] (2025).
- ²⁴K. Kim, S.-J. Baek, and H. J. Sung, "An implicit velocity decoupling procedure for the incompressible Navier–Stokes equations," *Int. J. Numer. Meth. Fluids* **38**, 125–138 (2002).
- ²⁵J. B. Perot, "An analysis of the fractional step method," *J. Comput. Phys.* **108**, 51–58 (1993).
- ²⁶W.-X. Huang, C. B. Chang, and H. J. Sung, "An improved penalty immersed boundary method for fluid–flexible body interaction," *J. Comput. Phys.* **230**, 5061–5079 (2011).
- ²⁷S. Hoyas and J. Jiménez, "Scaling of the velocity fluctuations in turbulent channels up to $Re_\tau = 2003$," *Phys. Fluids* **18**, 011702 (2006).
- ²⁸M. Bernardini and S. Pirozzoli, "Inner/outer layer interactions in turbulent boundary layers: A refined measure for the large-scale amplitude modulation mechanism," *Phys. Fluids* **23**, 061701 (2011).